

Detecting Ransomware Before Encryption: AI-Based Network Anomaly Detection

Balancing Automated Detection with AI Governance

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AI for Cybersecurity, Module 1 Assignment

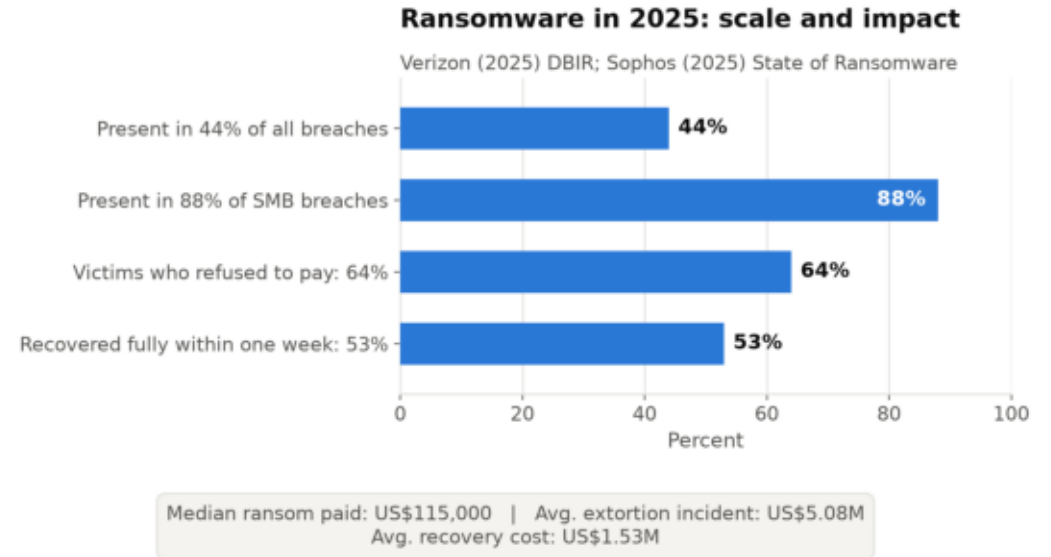
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Ransomware: Present in 44% of Data Breaches

- Ransomware is now present in 44% of data breaches, up 37% year-over-year (Verizon, 2025).
- Median ransom payments reached US\$115,000; still, 64% of victims refused to pay (Verizon, 2025).
- Average cost of an extortion or ransomware incident reaches \$5.08 million when disclosed by the attacker (IBM, 2025).
- This presentation examines whether AI-based anomaly detection can mitigate ransomware, and where it falls short.

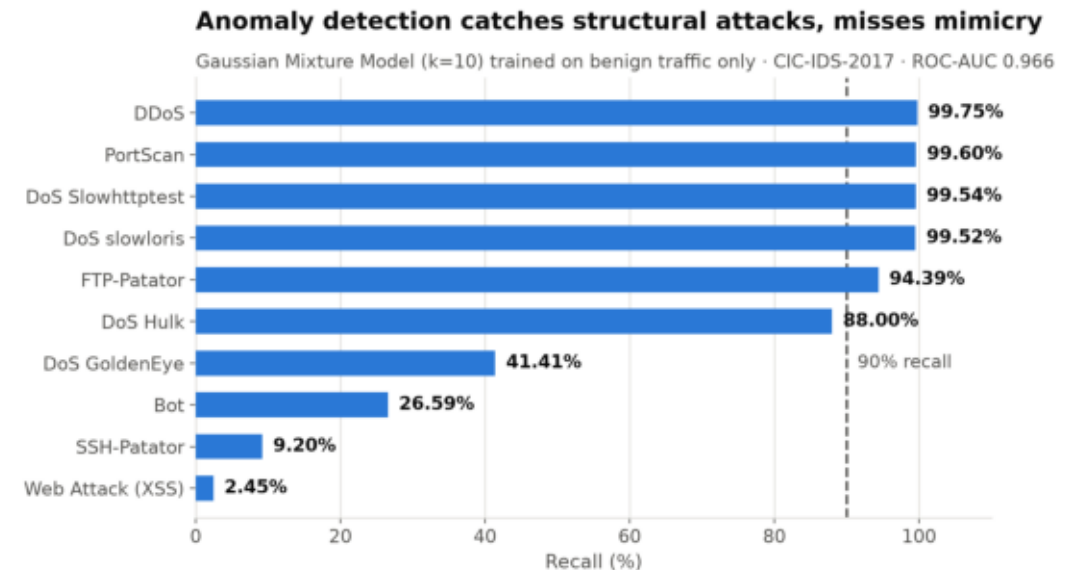
How Ransomware Escalates and What It Costs

- Ransomware appears in 44% of breaches overall, and 88% of breaches at small and medium businesses (Verizon, 2025).
- Attacks rose 37% year-over-year; median ransom payment reached US\$115,000 (Verizon, 2025).
- The kill chain includes command-and-control beaconing prior to deployment and lateral movement flagged via abnormal SMB traffic (Cybersecurity and Infrastructure Security Agency [CISA] et al., 2023), both occurring before encryption.



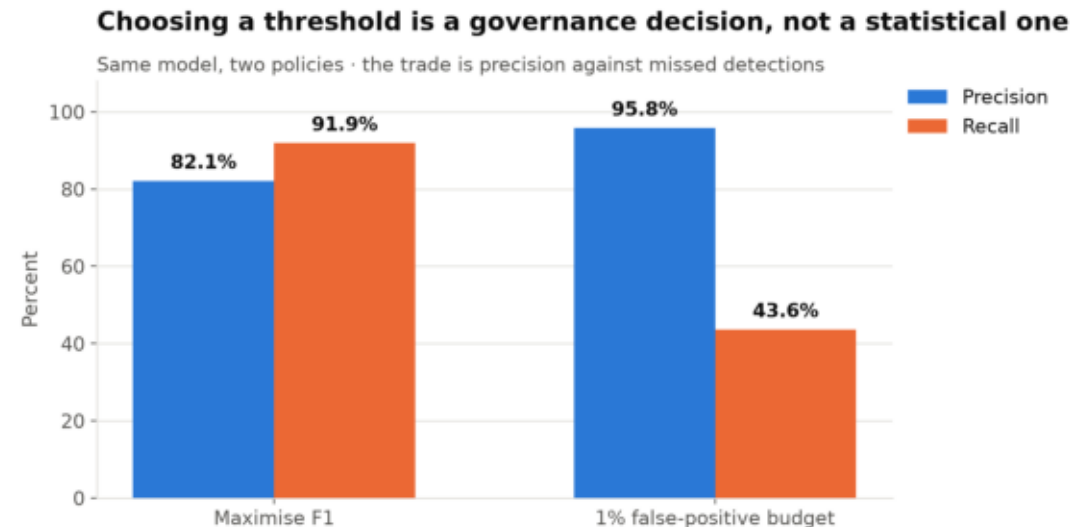
Network Anomaly Detection: Catching Ransomware Before Encryption

- Parisi (2019, ch. 5): model normal network behavior statistically (Gaussian density, clustering); flag deviation from that baseline as anomalous.
- C2 beaconing and lateral movement are network-visible deviations from baseline that occur before encryption (CISA et al., 2023).
- My lab: a Gaussian Mixture Model ($k=10$), trained on 1,243,352 benign flows only, reached ROC-AUC 0.9655 on 627,314 held-out flows.
- Near-perfect recall on structural attacks (PortScan 99.60%, DDoS 99.75%) but weak recall where traffic mimics normal sessions: SSH-Patator 9.20%, Bot 26.59%.
- The model catches some attack stages reliably and misses others entirely.



AI Governance: What Makes Automated Response Safe

- An anomaly-detection IDS makes automated, consequential decisions (blocking traffic, quarantining hosts) at a chosen false-positive budget.
- My lab: a 1% false-positive budget yields 95.8% precision at 43.6% recall, a governance decision about acceptable harm.
- 63% of breached organizations either lack an AI governance policy or are still developing one (IBM, 2025).
- One in five organizations reported a breach due to shadow AI, and only 37% have policies to manage AI or detect it (IBM, 2025).
- NIST's AI Risk Management Framework supplies the thresholds, auditability, and oversight needed before autonomous response is trustworthy (National Institute of Standards and Technology, 2023).



Deploying Anomaly Detection Under AI Governance

- Anomaly detection is most useful as an early-warning layer that still depends on a defined false-positive budget and a person to authorize action beyond that threshold.
- My own lab work, comparing a single Gaussian model to a Gaussian Mixture Model, taught me that model choice and threshold selection are inseparable, risk-based governance decisions.
- Any future deployment of this model would need a documented threshold, an audit trail, and a named person accountable for what it flags.

References

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